Unit I: Database Management and Web Development basics

Session 6 :

Triggers:

```
mysql> CREATE TABLE Driver (
    ->     driver_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     name VARCHAR(100),
    ->     email VARCHAR(100),
    ->     salary DECIMAL(10,2),
    ->     otp_verified BOOLEAN DEFAULT FALSE,
    ->     joining_date DATE,
    ->     last_login DATETIME
    -> );
ERROR 1046 (3D000): No database selected
mysql>
mysql> CREATE TABLE Driver_Audit (
    ->     audit_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     driver_id INT,
    ->     old_salary DECIMAL(10,2),
    ->     new_salary DECIMAL(10,2),
    ->     changed_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
    -> );
ERROR 1046 (3D000): No database selected
mysql> CREATE DATABASE lab_session;
Query OK, 1 row affected (0.01 sec)

mysql> USE lab_session;
Database changed
mysql> CREATE TABLE Driver (
    ->     driver_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     name VARCHAR(100),
    ->     email VARCHAR(100),
    ->     salary DECIMAL(10,2),
    ->     otp_verified BOOLEAN DEFAULT FALSE,
    ->     joining_date DATE,
    ->     last_login DATETIME
    -> );
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> CREATE TABLE Driver_Audit (
    ->     audit_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     driver_id INT,
    ->     old_salary DECIMAL(10,2),
    ->     new_salary DECIMAL(10,2),
    ->     changed_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
    -> );
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER //
mysql> CREATE TRIGGER PreventNegativeSalary
    -> BEFORE INSERT ON Driver
    -> FOR EACH ROW
    -> BEGIN
    ->     IF NEW.salary < 0 THEN
    ->         SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Error: Salary cannot be negative';
    ->     END IF;
    -> END //
Query OK, 0 rows affected (0.00 sec)
```

Working with Date:

Session 7

```
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER //
mysql> CREATE TRIGGER PreventNegativeSalary
    -> BEFORE INSERT ON Driver
    -> FOR EACH ROW
    -> BEGIN
    ->     IF NEW.salary < 0 THEN
    ->         SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Error: Salary cannot be negative';
    ->     END IF;
    -> END //
Query OK, 0 rows affected (0.00 sec)

mysql> DELIMITER ;
mysql> DELIMITER //
mysql> CREATE TRIGGER SalaryChangeAudit
    -> AFTER UPDATE ON Driver
    -> FOR EACH ROW
    -> BEGIN
    ->     IF OLD.salary <> NEW.salary THEN
    ->         INSERT INTO Driver_Audit(driver_id, old_salary, new_salary)
    ->         VALUES (OLD.driver_id, OLD.salary, NEW.salary);
    ->     END IF;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql> INSERT INTO Driver (name, salary) VALUES ('Failure Test', -1000);
ERROR 1644 (45000): Error: Salary cannot be negative
mysql> INSERT INTO Driver (name, email, salary, joining_date, last_login)
    -> VALUES ('Hasif', 'hasif@vtu.edu', 50000.00, '2026-02-01', NOW());
Query OK, 1 row affected (0.00 sec)

mysql> SELECT name, TIMESTAMPDIFF(DAY, joining_date, CURDATE()) AS Days_Since_Joined FROM Driver;
+-------+-------------------+
| name  | Days_Since_Joined |
+-------+-------------------+
| Hasif |                 2 |
+-------+-------------------+
1 row in set (0.00 sec)

mysql> SELECT CURDATE() AS Today, DATE_ADD(CURDATE(), INTERVAL 30 DAY) AS Expiry_Date;
+------------+------------+
| Today      | Expiry_Date |
+------------+------------+
| 2026-02-03 | 2026-03-05  |
+------------+------------+
1 row in set (0.00 sec)

mysql> SELECT name, joining_date FROM Driver WHERE DAYOFWEEK(joining_date) IN (1, 7);
+-------+--------------+
| name  | joining_date |
+-------+--------------+
| Hasif | 2026-02-01   |
+-------+--------------+
1 row in set (0.00 sec)
```

Indexes and Views:

Session 8:

```
+--------+--------------+
1 row in set (0.00 sec)

mysql> SELECT MONTHNAME(STR_TO_DATE('06-01-2006', '%d-%m-%Y')) AS Output_Month;
+--------------+
| Output_Month |
+--------------+
| January      |
+--------------+
1 row in set (0.00 sec)

mysql> CREATE INDEX idx_driver_email ON Driver(email);
Query OK, 0 rows affected (0.01 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> CREATE UNIQUE INDEX idx_unique_name ON Driver(name);
Query OK, 0 rows affected (0.01 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> CREATE INDEX idx_sal_date ON Driver(salary, joining_date);
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> SHOW INDEX FROM Driver;
+--------+------------+------------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+-------
-----+
| Table  | Non_unique | Key_name         | Seq_in_index | Column_name  | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment | Visible | Expres
sion |
+--------+------------+------------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+-------
-----+
| Driver |          0 | PRIMARY          |            1 | driver_id    | A         |           1 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL
     |
| Driver |          0 | idx_unique_name  |            1 | name         | A         |           1 |     NULL |   NULL | YES  | BTREE      |         |               | YES     | NULL
     |
| Driver |          1 | idx_driver_email |            1 | email        | A         |           1 |     NULL |   NULL | YES  | BTREE      |         |               | YES     | NULL
     |
| Driver |          1 | idx_sal_date     |            1 | salary       | A         |           1 |     NULL |   NULL | YES  | BTREE      |         |               | YES     | NULL
     |
| Driver |          1 | idx_sal_date     |            2 | joining_date | A         |           1 |     NULL |   NULL | YES  | BTREE      |         |               | YES     | NULL
     |
+--------+------------+------------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+-------
-----+
5 rows in set (0.01 sec)

mysql> DROP INDEX idx_driver_email ON Driver;
Query OK, 0 rows affected (0.01 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> SELECT name, CONCAT(SUBSTRING_INDEX(email, '@', 1), '123') AS Default_Password FROM Driver;
+-------+------------------+
| name  | Default_Password |
+-------+------------------+
| Hasif | hasif123         |
+-------+------------------+
1 row in set (0.00 sec)
```

SQL Built-in Functions:

Session 9

```
-----+
5 rows in set (0.01 sec)

mysql> DROP INDEX idx_driver_email ON Driver;
Query OK, 0 rows affected (0.01 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> SELECT name, CONCAT(SUBSTRING_INDEX(email, '@', 1), '123') AS Default_Password FROM Driver;
+-------+------------------+
| name  | Default_Password |
+-------+------------------+
| Hasif | hasif123         |
+-------+------------------+
1 row in set (0.00 sec)

mysql> SELECT name, COALESCE(email, 'No Contact Info') AS Primary_Contact FROM Driver;
+-------+-----------------+
| name  | Primary_Contact |
+-------+-----------------+
| Hasif | hasif@vtu.edu   |
+-------+-----------------+
1 row in set (0.00 sec)

mysql> SELECT CONCAT(UPPER(SUBSTRING(name, 1, 1)), LOWER(SUBSTRING(name, 2))) AS Normalized_Name FROM Driver;
+-----------------+
| Normalized_Name |
+-----------------+
| Hasif           |
+-----------------+
1 row in set (0.00 sec)

mysql> DELIMITER //
mysql> CREATE FUNCTION GetNetSalary(val DECIMAL(10,2)) RETURNS DECIMAL(10,2) DETERMINISTIC
    -> BEGIN
    ->     RETURN val - (val * 0.10);
    -> END //
Query OK, 0 rows affected (0.00 sec)

mysql> DELIMITER ;
mysql> DELIMITER //
mysql> CREATE FUNCTION GetDriverCategory(yrs INT) RETURNS VARCHAR(20) DETERMINISTIC
    -> BEGIN
    ->     IF yrs < 2 THEN RETURN 'Fresher';
    ->     ELSEIF yrs <= 5 THEN RETURN 'Junior';
    ->     ELSEIF yrs <= 10 THEN RETURN 'Mid';
    ->     ELSE RETURN 'Senior';
    ->     END IF;
    -> END //
Query OK, 0 rows affected (0.00 sec)

mysql> DELIMITER ;
mysql> SELECT
    ->     name,
    ->     salary,
    ->     GetNetSalary(salary) AS Take_Home_Salary,
    ->     GetDriverCategory(3) AS Experience_Level
```

User-Defined Functions:

Session 10

```
+-----------------+
1 row in set (0.00 sec)

mysql> DELIMITER //
mysql> CREATE FUNCTION GetNetSalary(val DECIMAL(10,2)) RETURNS DECIMAL(10,2) DETERMINISTIC
    -> BEGIN
    ->     RETURN val - (val * 0.10);
    -> END //
Query OK, 0 rows affected (0.00 sec)

mysql> DELIMITER ;
mysql> DELIMITER //
mysql> CREATE FUNCTION GetDriverCategory(yrs INT) RETURNS VARCHAR(20) DETERMINISTIC
    -> BEGIN
    ->     IF yrs < 2 THEN RETURN 'Fresher';
    ->     ELSEIF yrs <= 5 THEN RETURN 'Junior';
    ->     ELSEIF yrs <= 10 THEN RETURN 'Mid';
    ->     ELSE RETURN 'Senior';
    ->     END IF;
    -> END //
Query OK, 0 rows affected (0.00 sec)

mysql> DELIMITER ;
mysql> SELECT
    ->     name,
    ->     salary,
    ->     GetNetSalary(salary) AS Take_Home_Salary,
    ->     GetDriverCategory(3) AS Experience_Level
    -> FROM Driver;
+-------+----------+-----------------+------------------+
| name  | salary   | Take_Home_Salary | Experience_Level |
+-------+----------+-----------------+------------------+
| Hasif | 50000.00 |         45000.00 | Junior           |
+-------+----------+-----------------+------------------+
1 row in set (0.00 sec)

mysql> -- Trigger the update
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE Driver SET salary = 60000.00 WHERE name = 'Hasif';
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> -- See the audit trail
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT * FROM Driver_Audit;
+----------+----------+------------+------------+---------------------+
| audit_id | driver_id | old_salary | new_salary | changed_at          |
+----------+----------+------------+------------+---------------------+
|        1 |         1 |   50000.00 |   60000.00 | 2026-02-03 09:38:30 |
+----------+----------+------------+------------+---------------------+
1 row in set (0.00 sec)

mysql>
```